Sabik Swanan

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Career Objective

Computer Science undergraduate at BRAC University seeking an opportunity in machine learning, data science or software development, where I can apply hands-on experience in building end-to-end ML pipelines, geospatial data analysis and full-stack development to solve real-world problems. Currently developing a hybrid ML/DL framework with LLM integration for riverbank erosion prediction in Bangladesh, and eager to contribute to a team that values analytical rigour, ownership and continuous learning.

Education

BRAC University, Dhaka, Bangladesh 2022 – Present

B.Sc. in Computer Science | CGPA: 2.74 / 4.00

Rajuk Uttara Model College, Dhaka, Bangladesh 2020

Higher Secondary Certificate (HSC) | GPA: 5.00 / 5.00

Rajuk Uttara Model College, Dhaka, Bangladesh 2018

Secondary School Certificate (SSC) | GPA: 4.94 / 5.00

Research Experience

A Hybrid ML/DL Framework with LLM Integration for Bangladesh-Wide Riverbank Erosion Prediction Using Multi-Temporal Satellite Data Ongoing

Undergraduate Thesis, BRAC University — Supervisor: Dr. Chowdhury Mofizur Rahman

- Developing an erosion-prediction framework for the Jamuna River near Sirajganj, using Random Forest as the primary model with U-Net and CNN-LSTM as supporting deep-learning components.
- Built an end-to-end satellite-data pipeline on Google Earth Engine in Google Colab, drawing on JRC Global Surface Water occurrence (1984–2021), and exported prediction maps as GeoTIFFs at 30 m and 60 m resolution.
- Identified and removed target leakage from a water-occurrence feature computed over the full 1984–2021 record, and corrected spatial drift in map rendering by clipping outputs to the study area.
- Added SHAP-based explainability and an LLM/RAG reporting layer that runs a locally hosted Qwen2.5-1.5B-Instruct model to generate plain-language erosion reports, with no dependence on paid external APIs.
- Completed the literature review (Phase 1) with review-panel feedback addressed, and the full implementation pipeline (Phase 2); Phase 3 in progress.

Projects

Flood Probability Prediction Using Machine Learning *Python, Scikit-learn, Pandas, Seaborn*

- Modelled flood probability from 20 environmental and socio-economic risk factors on a 50,000-record dataset, with correlation analysis, standard scaling and an 80/20 train–test split.
- Compared SVR, Decision Tree, Random Forest and KNN regressors; KNN performed best (R^2 0.777, RMSE 0.024, MAE 0.019) ahead of Random Forest (R^2 0.732).

Travel & Tourism Website (Team Project) *React, Node.js, Express, MongoDB, Tailwind CSS*

- Full-stack travel-package platform with JWT and bcrypt authentication, a separate admin view, Nodemailer email notifications and Google Gemini API integration; deployed on Vercel (source).
- Interactive Leaflet/OpenStreetMap maps that detect destinations from each package itinerary and plot them automatically, with no paid map API.

Tank Battle — 3D Game *Python, PyOpenGL, GLUT*

- 3D tank-defence game with arcing projectile physics, progressively harder enemy waves, power-ups, animated HP bars, orbit and follow cameras, a minimap, scoring, and pause/restart.

Smart Transit Network System (Team Project) *Cisco Packet Tracer*

- Designed and simulated a multi-site transit network with VLSM subnetting and static and dynamic routing; responsible for router configuration and network services.

Skills

Programming	Python, SQL
ML & AI	TensorFlow, Scikit-learn, XGBoost, MNE-Python, SHAP, Prompt Engineering
Geospatial & Web	Google Earth Engine, PyOpenGL, Cisco Packet Tracer
Databases	MongoDB, PostgreSQL, MySQL, Oracle
Tools & Hosting	Figma, Canva, Looker Studio, Microsoft Office, Google Workspace, Vercel

Key Skills

- Stakeholder communication
- Administrative support & reporting
- Data & record management
- Scheduling, follow-ups & coordination

Soft Skills

- Analytical mindset and attention to detail
- Clear communicator across technical and non-technical contexts
- Team-oriented with ownership and accountability
- Adaptable, ethical and growth-focused

Achievements, Certifications & Activities

The Duke of Edinburgh's International Award — Bronze Standard Mar 2023

The Duke of Edinburgh's International Award Foundation | Associated with BRAC University

- Completed a structured personal-development programme of community service, physical activity, skill development and an adventurous journey.

Competitive Programming — Codeforces (SabikSwanan)

- Solved 63 algorithmic problems in Python, covering graph algorithms (BFS, Dijkstra's shortest path, topological sort, tree diameter), minimum spanning trees with Kruskal's algorithm and Union-Find, and greedy scheduling.

Assistant Director — BRAC University Business Club (BIZ BEE) Apr 2023 – Jan 2024

BRAC University

Volunteer — International Coastal Cleanup Nov 2023

Kewkradong Bangladesh, Ocean Conservancy & The Coca-Cola Company | Cox's Bazar

- Took part in a two-day beach cleanup carried out during Cyclone Midhili.

Certifications

- Reshaping Data with pandas — DataCamp
- Cleaning Data in Python — DataCamp
- Introduction to Importing Data in Python — DataCamp
- Intermediate Importing Data in Python — DataCamp

Language Proficiency

- **English:** Advanced
- **Bengali:** Native
- **Spanish:** Beginner